

Title: Brazil's president pledges to hold dialogue with protesters
Description: SAO PAULO/BRASILIA (Reuters) - Brazilian President Dilma Rousseff promised on Friday to hold a dialogue with members of a protest movement sweeping the country, but also said she would do whatever is necessary to maintain order in the wake of widespread vandalism and looting.

Title: U.S. files criminal charges against Snowden over leaks - U.S. sources
Description: WASHINGTON (Reuters) - The U.S. government has filed sealed criminal charges against former National Security Agency contractor Edward Snowden, who admitted leaking secrets about classified U.S. surveillance programs, U.S. sources said on Friday.

Box 1

SAO PAULO/BRASILIA (Reuters) - Brazilian President Dilma Rousseff promised on Friday to hold a dialogue with members of a protest movement sweeping the country, but also said she would do whatever is necessary to maintain order in the wake of widespread vandalism and looting.

WASHINGTON (Reuters) - The U.S. government has filed sealed criminal charges against former National Security Agency contractor Edward Snowden, who admitted leaking secrets about classified U.S. surveillance programs, U.S. sources said on Friday.

Box 2

```
$ apt-get install rsstail
```

rsstail will get installed properly without the need for specifying any other detail. It is assumed that you have direct Internet access via your home router, with no proxy involved. If a proxy is present that requires credentials/authentication; specify the proxy in `/etc/apt/apt.conf.`

espeak. It is a text-to-speech utility available on most Linux distributions. A few other utilities such as festival are also available. I installed both festival and espeak but found espeak to be more configurable.

To install espeak, type the below-given command as a root user on Raspberry Pi:

```
$ apt-get install espeak
```

alsa-utils. alsa-utils is the package that enables sound on Raspberry Pi. In case alsa-utils is not pre-installed on your Raspberry Pi, try installing it with the command:

```
$ apt-get install alsa-utils
```

That is all you need to install or update to get text-to-speech conversion working on your Raspberry Pi. For a detailed set of instructions and variations, you can refer to the article at [http://elinux.org/RPI_Text_to_Speech_\(Speech_Synthesis\)](http://elinux.org/RPI_Text_to_Speech_(Speech_Synthesis))

Step 3: Fetch and format RSS feed articles. RSS feed is a very simple

means of accessing information from any source that supports RSS feeds. The feed comprises a predefined structure/format like title, description, heading and published time for every article, which makes it easy and programmable for others to subscribe to such feed and scrap out the required element per article and do the required processing. For example, in this project, we are mainly interested in just the description part of the RSS feed article. Since RSS is an established standard, you can program to receive RSS feeds from a variety of sources without worrying about the syntax part of the feed.

Let us first try to get the RSS content and print it on the screen by typing the command:

```
$ rsstail -i 10 -u http://feeds.reuters.com/reuters/INtopNews -d -H -1
```

The output should be something like Box 1.

As you can see, for every news item 'Title' and 'Description' elements are printed out. To understand the various options used in rsstail command, you can refer the manual by typing 'man rsstail' on your command prompt.

If needed, you can even weed out some of the unwanted elements and format the output from rsstail command below to just get description for every news item on a new line (see Box 2).

```
$ rsstail -i 10 -u http://feeds.reuters.com/reuters/INtopNews -d -H -1 | grep Description | sed -e 's/Description/'
```

Step 4: Converting RSS feed to speech. The last step is to feed the selected output from rsstail into your text-to-speech utility (espeak). In UNIX systems this is very simple and easy as all you have to do is to pipe the output next to the espeak command utility. Connect a local speaker to the audio jack of Raspberry Pi and type the following command to listen to the output:

```
$ rsstail -i 10 -u http://feeds.reuters.com/reuters/INtopNews -d -H -1 | grep Description | head $3 | sed -e 's/Description:/' | espeak -ven+f3 -k5 -a130 -a200 2>/dev/null
```

After you successfully listen to your RSS feed, replace your local speaker and insert the FM transmitter's input cable in Raspberry Pi's audio jack. Switch on the FM transmitter and tune your radio to the frequency at which your FM transmitter is broadcasting the RSS feeds.

Step 5: Wrapping up and scheduling the script. So far we have run all tasks one at a time—from connecting to RSS source on the Internet, to fetching the content, filtering and formatting it, and converting to speech. Now it is time to neatly wrap these up in various handy scripts and schedule so that your very own personal FM radio station becomes more automated.

Create a file with the name 'rssFeed.lst' to store the various RSS source URLs that you can crawl and use as source (see Box 3).

Create another file called 'rssFetch.sh' to wrap up the single command line that prints the RSS content on the screen (see Box 4).

Save the file and set its permission to executable by using command:

```
$ chmod 755 rssFetch.sh
```

Next, create a wrapper script (rssTTS.sh) to pick up one URL source at a time from 'rssFeed.lst' file and use 'rssFetch.sh' script to emit the RSS news article in speech format (see Box 5).

Save the file and set its permission